Day-12 Task-3: Schema design and querying on insurance claims dataset (Day-10 Task-5 dataset)

1. Database creation

```
CREATE DATABASE DB_DAY_12_TASK_2;
USE DB_DAY_12_TASK_2;
```

2. Schema creation (table)

```
CREATE TABLE InsuranceClaims(
    customer_id VARCHAR(16) NOT NULL,
    age INTEGER NOT NULL CHECK (age > 18),
    annual_income FLOAT NOT NULL CHECK (annual_income >= 0),
    insurance_type VARCHAR(32) NOT NULL,
    premium_amount FLOAT NOT NULL CHECK (premium_amount >= 2000),
    claim_amount FLOAT NOT NULL CHECK (claim_amount >= 0),
    policy_status VARCHAR(16)
);
```

3. Data Insertion

```
INSERT INTO InsuranceClaims VALUES ('C-1001', 25, 350000, 'Health', 12000,
0, 'Active');

INSERT INTO InsuranceClaims VALUES ('C-1002', 42, 850000, 'Life', 18000,
50000, 'Active');

INSERT INTO InsuranceClaims VALUES ('C-1003', 36, 620000, 'Vehicle',
15000, 22000, 'Active');

INSERT INTO InsuranceClaims VALUES ('C-1004', 55, 1100000, 'Health',
24000, 175000, 'Expired');

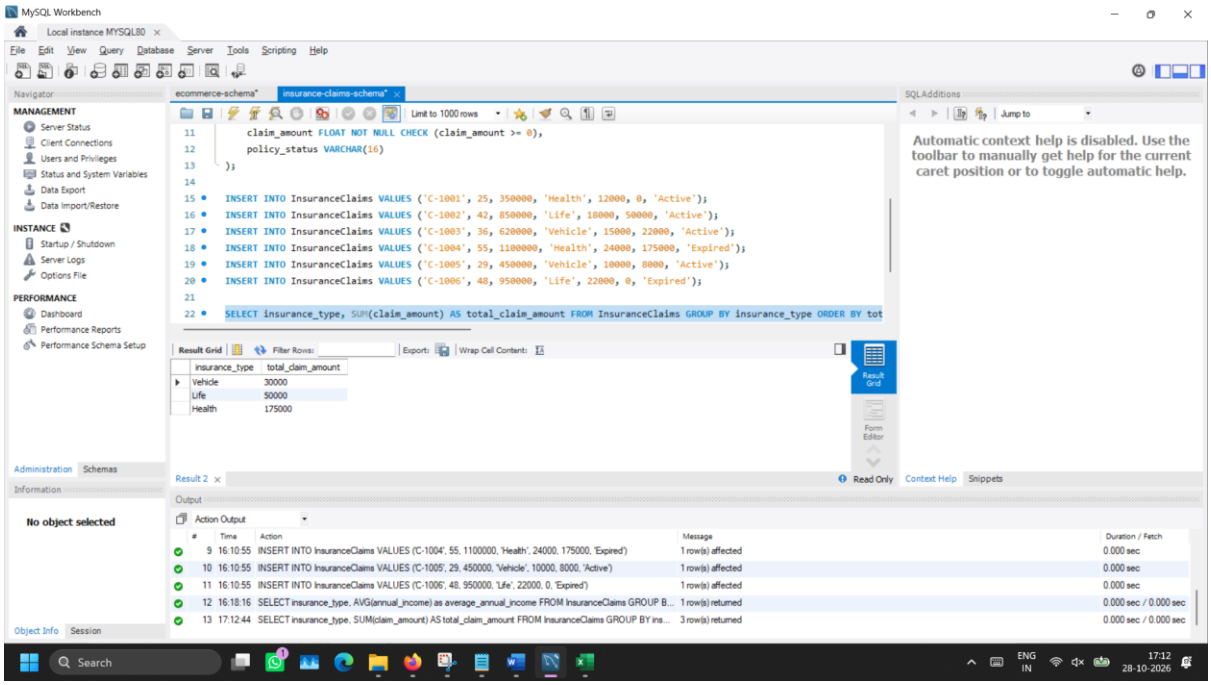
INSERT INTO InsuranceClaims VALUES ('C-1005', 29, 450000, 'Vehicle',
10000, 8000, 'Active');

INSERT INTO InsuranceClaims VALUES ('C-1006', 48, 950000, 'Life', 22000,
0, 'Expired');
```

4. Grouping, Sorting and applying aggregate function

Display total claim amount of each type of insurance in ascending order

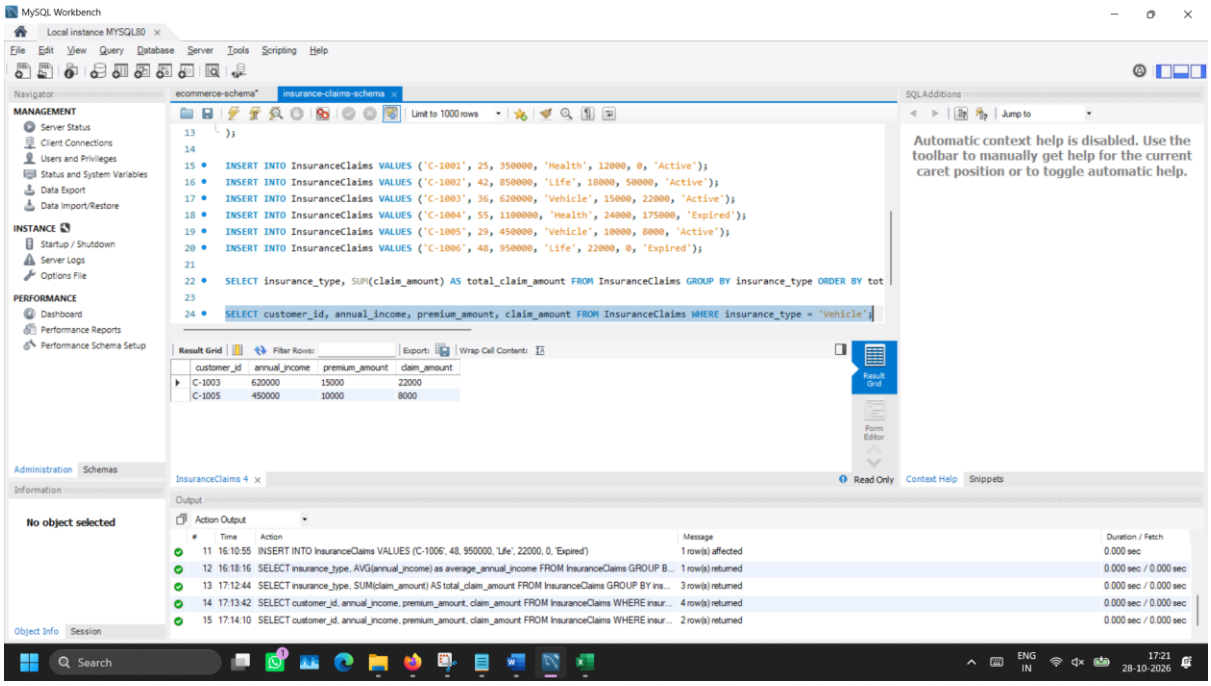
```
SELECT insurance_type, SUM(claim_amount) AS total_claim_amount FROM
InsuranceClaims GROUP BY insurance_type ORDER BY total_claim_amount;
```



5. WHERE clause usage

Retrieve insurance claims info which belongs to Vehicle insurance type.

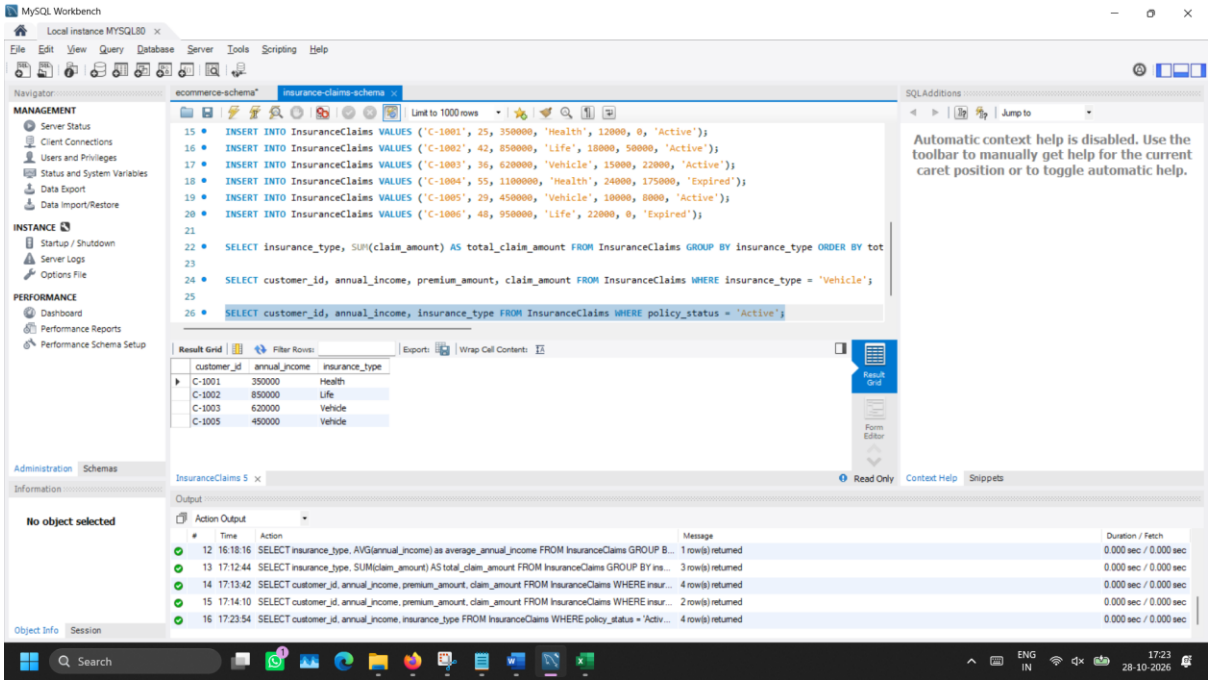
```
SELECT customer_id, annual_income, premium_amount, claim_amount FROM
InsuranceClaims WHERE insurance_type = 'Vehicle';
```



6. Applying Filter

Identify all active insurance claims.

```
SELECT customer_id, annual_income, insurance_type FROM InsuranceClaims
WHERE policy_status = 'Active';
```



7. Grouping, average and identifying top insurance type

Find the insurance type which has highest average annual income.

```
SELECT insurance_type, AVG(annual_income) as average_annual_income FROM
InsuranceClaims GROUP BY insurance_type ORDER BY average_annual_income
DESC LIMIT 1;
```

